Piezoelectric Floor Tile — Design Iterations

Oct 31, 2026 · Wenhao Jia

Overview

This project builds a floor tile that turns footstep pressure into electricity with piezoelectric discs. It started in June 2026 by replicating the circuit from a published tile, then moved through three mechanical designs to raise output per step.

Item	Detail
Team	Shaoxuan Xiong, Shuoqi Yang, Wenhao Jia (listed in no particular order)
School	Queens High School for Language Studies, Engineering Research
Reference prototype	Selim, Yehia & Saleeb, Energy Harvesting Floor Tile Using Piezoelectric Patches for Low-Power Applications, J. Vib. Eng. Technol., 2024
Reference design	455 × 405 mm sandwich tile, 14 piezo patches in parallel, 1N60P bridge rectifier + 2.2 µF capacitor; 33.7 V peak open-circuit, 2 LEDs lit, 246 mW max useful power per tile

Development path:

1. Phase 0 — circuit replication (June 2026): rectifier bridge, capacitor, LED on a bench.
2. Version 1 — foam sandwich tile: piezo discs embedded in foam between two floor layers.
3. Version 2 — spring + impactor tile: springs and an impactor strike the piezo.
4. Version 3 — center-punch tile: a mechanism modeled on an automatic center punch.
5. Drop-test rig: controlled free-fall tests of impact variables.

Phase 0 — Circuit replication (June 2026)

The team rebuilt the reference paper's harvesting circuit and lit an LED on June 12, after finding that the original storage capacitor was far too large.

Target circuit: piezo discs in parallel → bridge rectifier (4 × 1N60P diodes) → storage capacitor → LED.

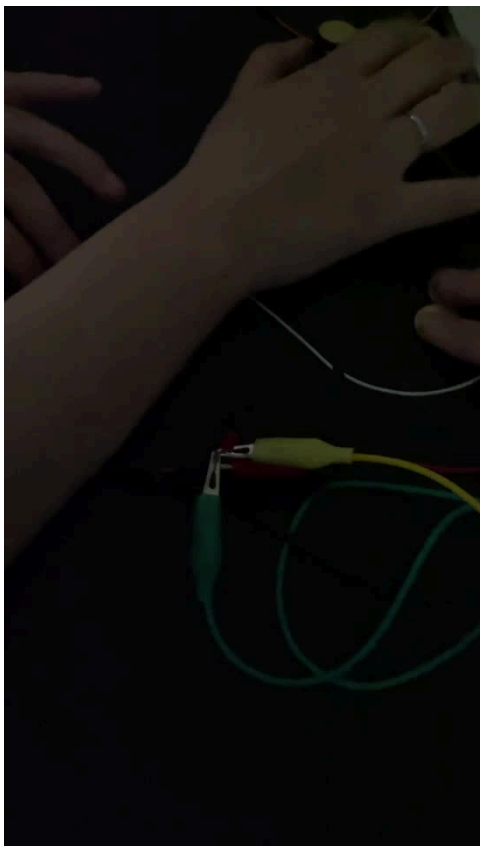
Lab log

Date	Work done	Result	Problem found
Jun 1	Built the bridge on a breadboard with 100 µF / 25 V capacitor, 150 Ω resistor, yellow LED	Failed: LED did not light	Possible miswired diode node or breadboard row; 100 µF + 150 Ω drains charge too fast; only 4–5 discs vs. 14 in the paper; multimeter too slow for ms-scale pulses
Jun 2	Team A (Xiong, Yang) soldered a 1N60P bridge; Team B (Jia) tried a similar circuit from another paper on a breadboard	Bridge passed multimeter test; breadboard LED circuit failed	Capacitor, resistor and LED not yet connected
Jun 4	Soldered 5 rectifier bridges; wired 14 piezo discs in parallel	All parts conduct (multimeter)	Unstable solder joints broke disc-to-trunk connections; no data recorded
Jun 11	Swapped thin jumpers for thick copper trunk wire; embedded 40 mm discs in foam base; added bridge + large capacitor	LED did not light	Cause unknown at the time
Jun 12	Wire comparison, full circuit with 5 discs, oscilloscope node measurements, capacitor sweep	LED lit with 2.2 µF capacitor	See below

June 12 node measurements (5 discs, oscilloscope)

Stage	Configuration	Voltage	LED (needs 1.8–2.2 V)
1	Piezo discs, direct output	~5 V	—
2	+ 1N60P bridge	~2 V	—
3	+ original large capacitor	~600 mV	Off
4	Bridge + 0.1 µF	≤ 508 mV	Off
5	Bridge + 2.2 µF (paper value)	2–3 V	On

Wire gauge made no difference to output. Two causes kept the LED off: voltage lost across the rectifier, and a capacitor so large that the small charge from tapping gave only ~600 mV ($V = Q / C$).



LED test: tapping the piezo discs lights the LED *(video — still frame; play it in the online doc)*

Lessons carried forward

- Match capacitor size to the charge per step; 2.2 µF worked for 5 discs.
- Use an oscilloscope, not a multimeter, for millisecond pulses (Time/div, Volt/div, edge trigger, DC coupling).
- Solder joints need strain relief; detachable piezo modules were planned.
- For a real floor (14 discs): body weight can crack discs, so a limit stop is needed; walking may deform discs too little, so series/parallel layout and a boost converter were proposed.

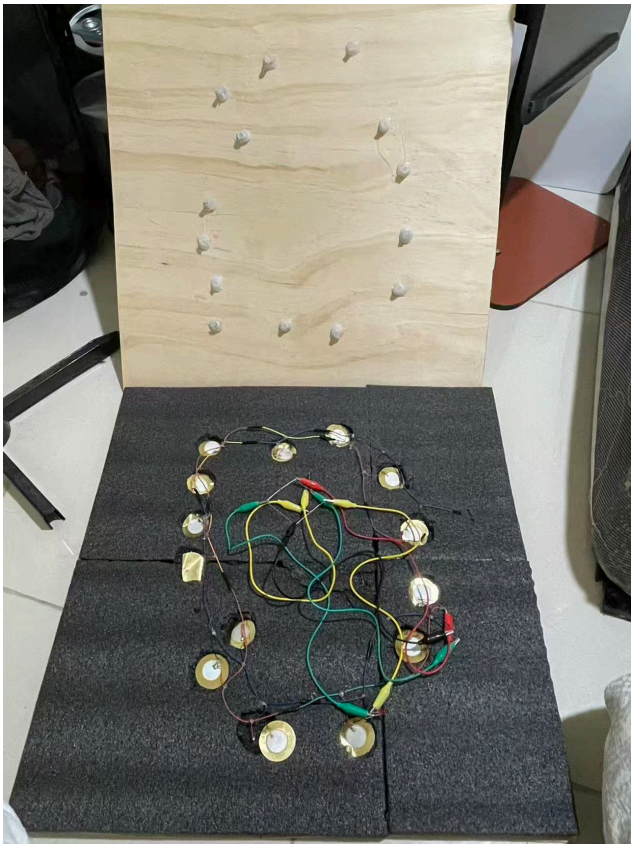
Version 1 — Foam sandwich tile

Version 1 copied the reference paper's sandwich tile directly; stepping on it cracked the piezo discs, because the rigid hot-melt glue sticks pressed straight into them. A second build replaced the glue sticks with soft foam pads.

1a — Direct replication (hot-melt glue sticks)

Layer	Material	Detail
Top plate	Plywood sheet	Short hot-melt glue sticks bonded to the underside, one above each disc, as in the paper
Middle layer	Black foam sheet, four pieces	Holes cut for about 14 piezo discs
Wiring	Disc leads + trunk wire, alligator clips	Discs in parallel; tested with a multimeter

Result: one step on the tile cracked the discs. The glue sticks are rigid, so the full load went into a small contact point on each disc with nothing to limit its deflection.



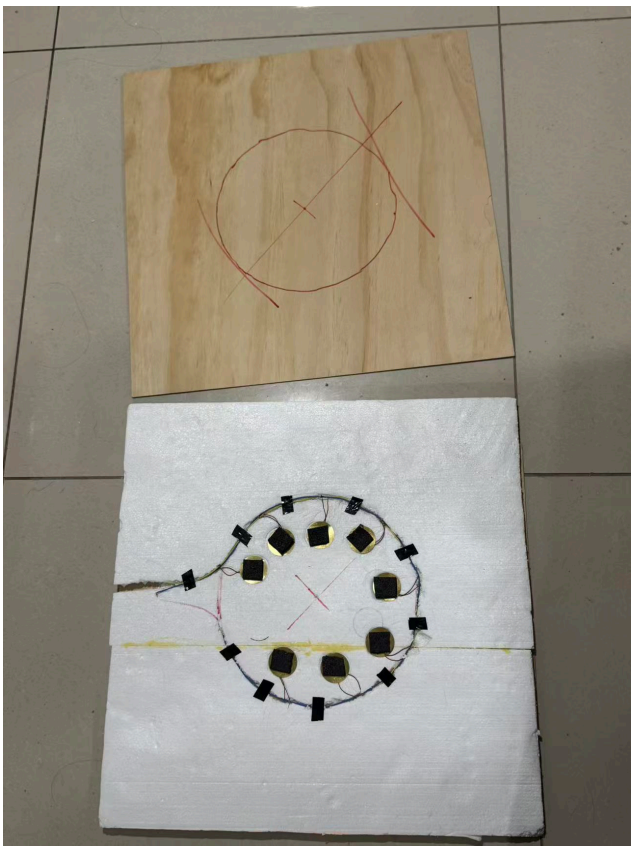
1a: plywood top plate with glue-stick contacts (top) and black foam layer with the disc layout (bottom)



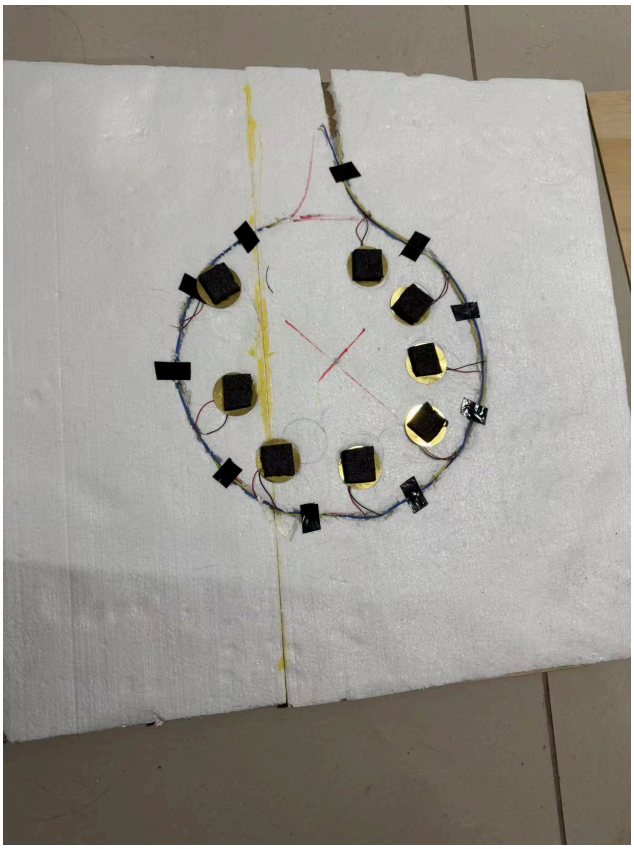
1a: two foam layers with disc layouts, multimeter testing on the right

1b — Foam pads instead of glue sticks

Layer	Material	Detail
Top plate	Plywood sheet	Circle and center lines drawn to align with the disc ring below
Middle layer	White foam board, two pieces glued along a seam	8 piezo discs set in a ring; wires run in a groove around the ring to one exit channel
Force pads	Black foam squares	One on each disc, to pass top-plate pressure onto the disc
Wiring	Thin disc leads + trunk wire	Discs in parallel on the ring trunk, held with black tape



Version 1: plywood top plate (top) and foam middle layer with the piezo ring (bottom)



Version 1: close-up of the 8-disc ring, foam pads and wiring groove

Testing

Open-circuit test (October 2026): one person stepped on the tile, 10 steps per set, 5 sets at the center and 5 at the edge. Piezo output went straight to a handheld oscilloscope (X10 probe, DC, 200 ms/div). Readings were taken from photos of the screen.

A step at the edge gives about 39 V peak-to-peak, roughly a quarter of the center output, so Version 1's output depends heavily on where the foot lands.

Center

The tile gives open-circuit swings of roughly 150–190 V peak-to-peak per step, but most readings hit the scope's screen limit, so the true peaks are higher. A clean average needs a retest at 50 V/div.

Set	Scale	Steps read	Clipped	Partial capture	Unclipped Vpp (V)
1	10 V/div	10	10	0	— (all ≥ 76)
2	20 V/div	9	5	1	136.0, 180.0, 181.6
3	20 V/div	9	7	0	149.6, ~180.4
4	20 V/div	8	4	1	160.8, 171.2, 176.8
5	20 V/div	10	3	1	158.4, 161.6, 161.6, 171.2, 173.2, 188.0

- **Clipped** = max reading at the screen limit (+50.8 V at 10 V/div, +101.6 V at 20 V/div) or min at −51.2 / −102.4 V. These are lower bounds only.
- **Partial capture** = only the negative pulse was on screen (max < 5 V, Vpp 60–78 V); left out of the range.
- Unreadable or duplicate photos were dropped: set 2 had one duplicate, set 3 one blurred photo, set 4 two.
- Across the 14 clean, unclipped steps, Vpp is 136–188 V (median about 171 V). This underestimates the tile because 29 of 46 steps clipped.
- Each waveform shows a press pulse and an opposite release pulse, followed by a short ringing tail.

Edge

All edge readings stayed on screen at 20 V/div, so these are full values.

Set	Steps used	Mean Vpp (V)	Range (V)
1	10	35.6	22.4–49.2
2	10	37.1	32.0–42.4
3	10	39.8	33.6–48.8
4	9	39.1	32.0–45.6
5	9	43.5	32.8–57.6
All	48	38.9	22.4–57.6

- Two steps (set 4 and set 5, first step each) caught only the negative pulse (max 0.8 V) and were left out.
- A few blurred photos were read from the on-screen Vpp; their max/min may be off by a few volts.

Center vs. edge

Position	Clean steps	Vpp (V)	Note
Center	14 of 46	136–188, median ~171	Lower bound: 29 steps clipped
Edge	48 of 50	22–58, mean 38.9	No clipping

The center gives at least about 4.4 times the edge Vpp; the true ratio is likely higher because the center readings are clipped.

Single-LED load, center

With one LED as the load, the measured voltage never rose above about 17.6 V, and after the first step each step swung it by about 10.7 V. Same method: center steps, 10 per set, 5 sets, 10 V/div (50 readings).

Set	First step Vpp (V)	Later steps: mean Vpp (V)	Later steps: min (V)
1	18.0	11.6	4.4–8.0
2	18.0	10.5	6.0–8.0
3	18.0	10.3	6.0–7.6
4	18.4	10.2	6.8–8.0
5	17.6	10.7	5.6–7.6
All	18.0	10.7	4.4–8.0

- The maximum sat at 17.2–18.0 V in all 50 readings, well inside the 10 V/div screen range, so a circuit element limits it, not the scope.

- Each set starts from about 0 V (first-step min −0.4 to −0.8 V); afterwards the voltage no longer returns to zero but stays at 4–8 V between steps.
- The trace is one-sided: a step up on each press, then a slow decay, like a capacitor being topped up and drained.

Likely reading (to confirm): after the rectifier, each step charges a storage capacitor to about 17.6 V; the LED drains it until the voltage falls below where the LED stops conducting. If so, the ~10.7 V swing reflects the charge passed through the LED on each step.

Open question: circuit (rectifier? capacitor value? series resistor?), probe points, and whether the LED lit on each step.

Single-LED load, edge

At the edge, each step after the first swung the voltage by about 8.3 V, and the maximum never reached the 17.6 V level seen at the center. Same method, 10 V/div (48 readings; sets 1 and 3 have 9).

Set	First step Vpp (V)	Later steps: mean Vpp (V)	Later steps: max (V)
1	7.6	7.9	10.8–14.4
2	8.0	8.0	12.0–15.6
3	8.4	8.2	12.0–15.6
4	11.2	9.0	14.0–16.8
5	14.8	8.2	11.6–15.2
All	10.0	8.3	10.8–16.8

- Between steps the voltage stayed at 3.6–7.2 V, close to the center's 4.4–8.0 V floor.
- The first-step charge grew from set to set (7.6 V to 14.8 V), so the start condition was not identical; later steps were steady.

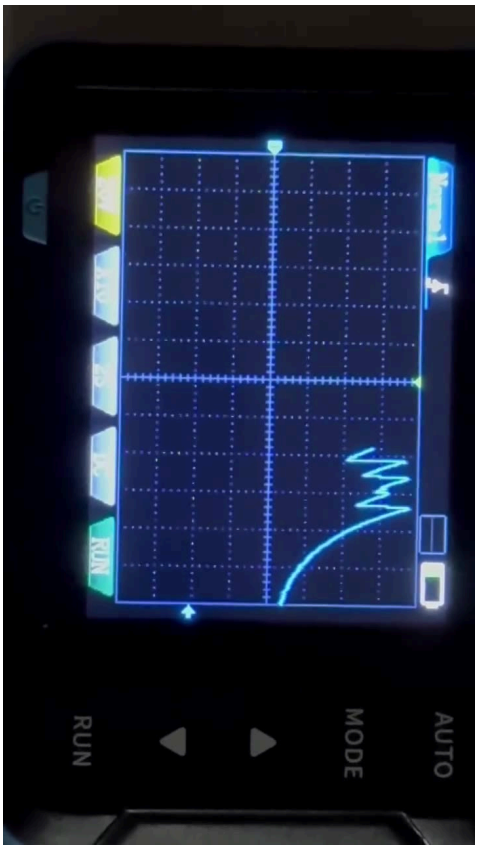
LED load: center vs. edge

Position	Later steps Vpp (V)	Max reached (V)	Open-circuit Vpp (V)
Center	10.7	17.2–18.0 (every step)	≥ 136–188
Edge	8.3	10.8–16.8	22–58

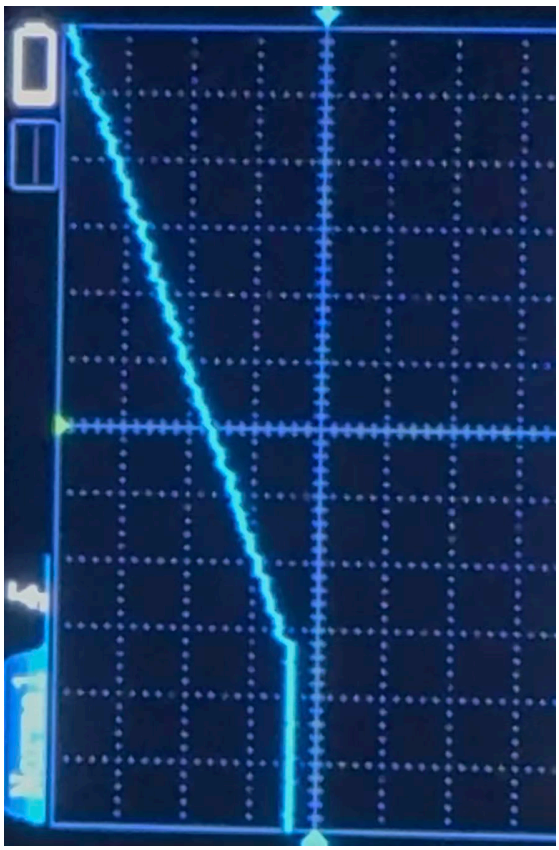
Under the LED load the center gives only about 1.3 times the edge, versus 4.4 times or more in open circuit. The likely reason is the ~17.6 V ceiling: center steps are cut off at it, so the load test hides most of the center's advantage. It is useful for showing the LED circuit works, not for comparing positions.

Capacitor charge and discharge

Recordings of the storage capacitor charging over repeated steps and discharging afterwards. Open question: capacitor value, number of steps, peak voltage reached and discharge path (LED or resistor).



Capacitor discharge recording (15 s) *(video — still frame; play it in the online doc)*



Capacitor charging trace on the oscilloscope

To fix: set 50 V/div, keep the whole trace on screen, and record max, min and Vpp for every step (CSV export if the scope supports it).

Open question: confirm that this was build 1b with no rectifier, and which edge point was stepped on (over a disc or between discs).

Lessons learned

Open question: what limited Version 1 and led to the spring + impactor design.

Version 2 — Spring + impactor tile

Version 2 adds springs between the top plate and the base so the plate travels a controlled distance under each step. The design was driven by a simulation built from measured step motion.

Step-motion measurement

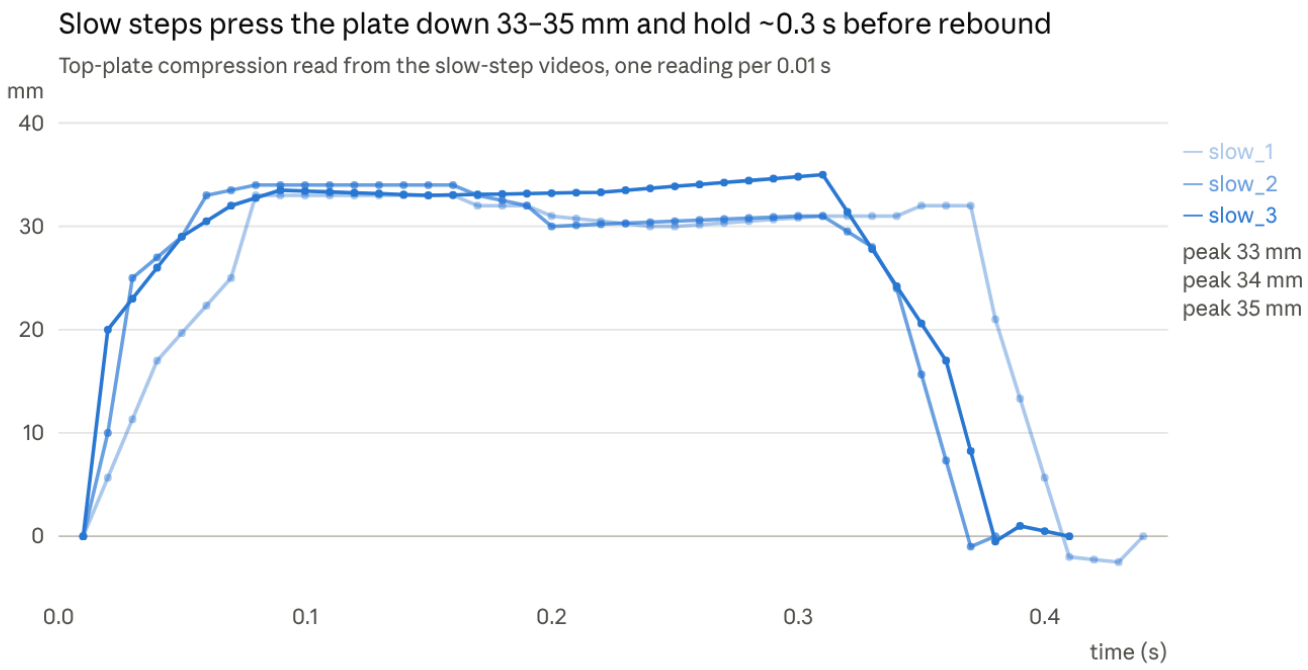
Top-plate displacement was filmed against a steel ruler while a person stepped slowly onto the plate. The footage gives plate travel over time, which was entered into Excel and used as input for the simulation.



Slow-step recording 1: plate travel against a ruler (27 s) *(video — still frame; play it in the online doc)*



Slow-step recording 2: plate travel against a ruler (36 s) *(video — still frame; play it in the online doc)*



Slow-step data sheet (3 trials, 0.01 s steps); gap measured from 106 mm unloaded

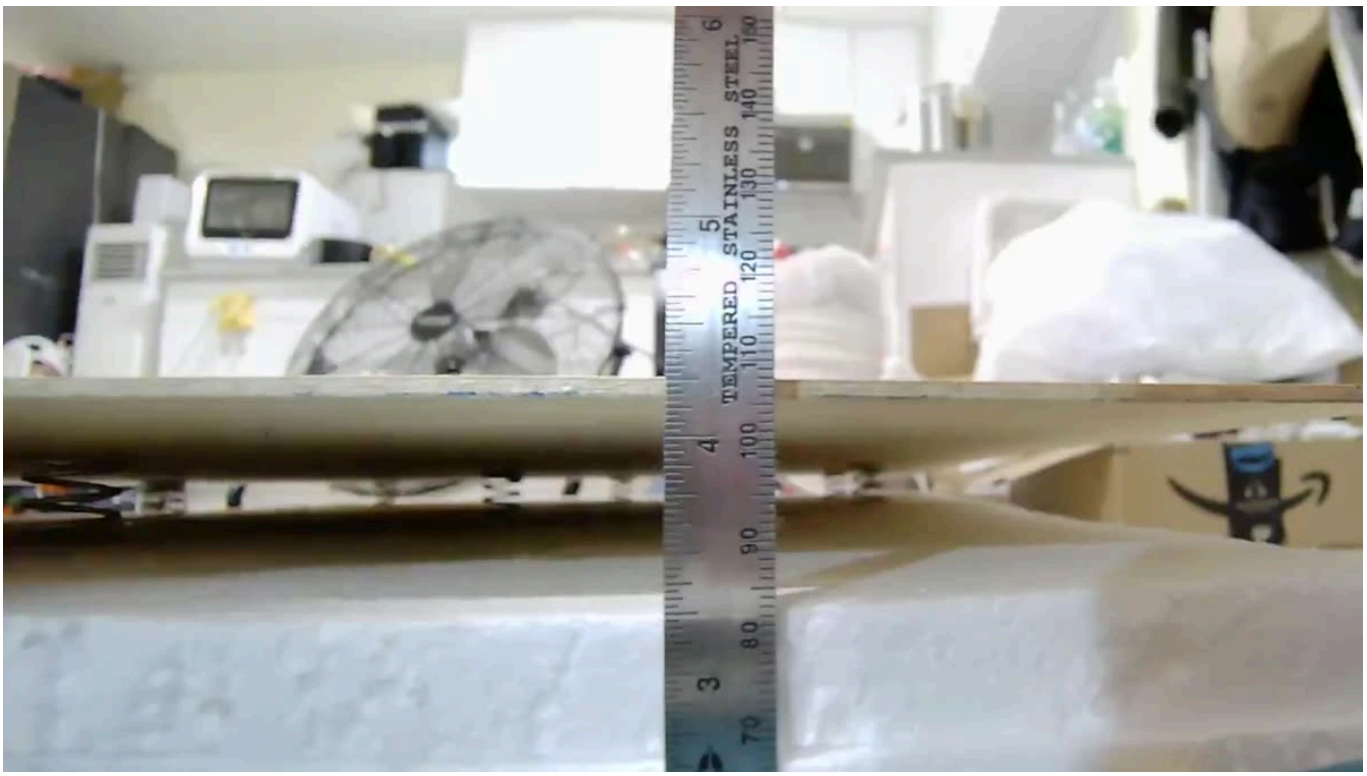
Slow-step data sheet (3 trials, 0.01 s steps); gap measured from 106 mm unloaded

The plate drops about 30 mm in the first 0.06 s, stays near its limit, then springs back to zero in roughly 0.06 s. The small negative values at the end (down to −2.5 mm) are likely reading or zero-point error; the simulation clamps them to 0. Three fast-step trials (fast_1–3, 0.31–0.32 s, peak 25–32 mm) were recorded the same way.

Source recordings for the simulation (July 1, 2026)

The six curves fed into the simulation came from these clips: three fast steps and three slow steps, filmed at 120 fps, 1920 × 1080. Clips are shown here downscaled to 1280 px.

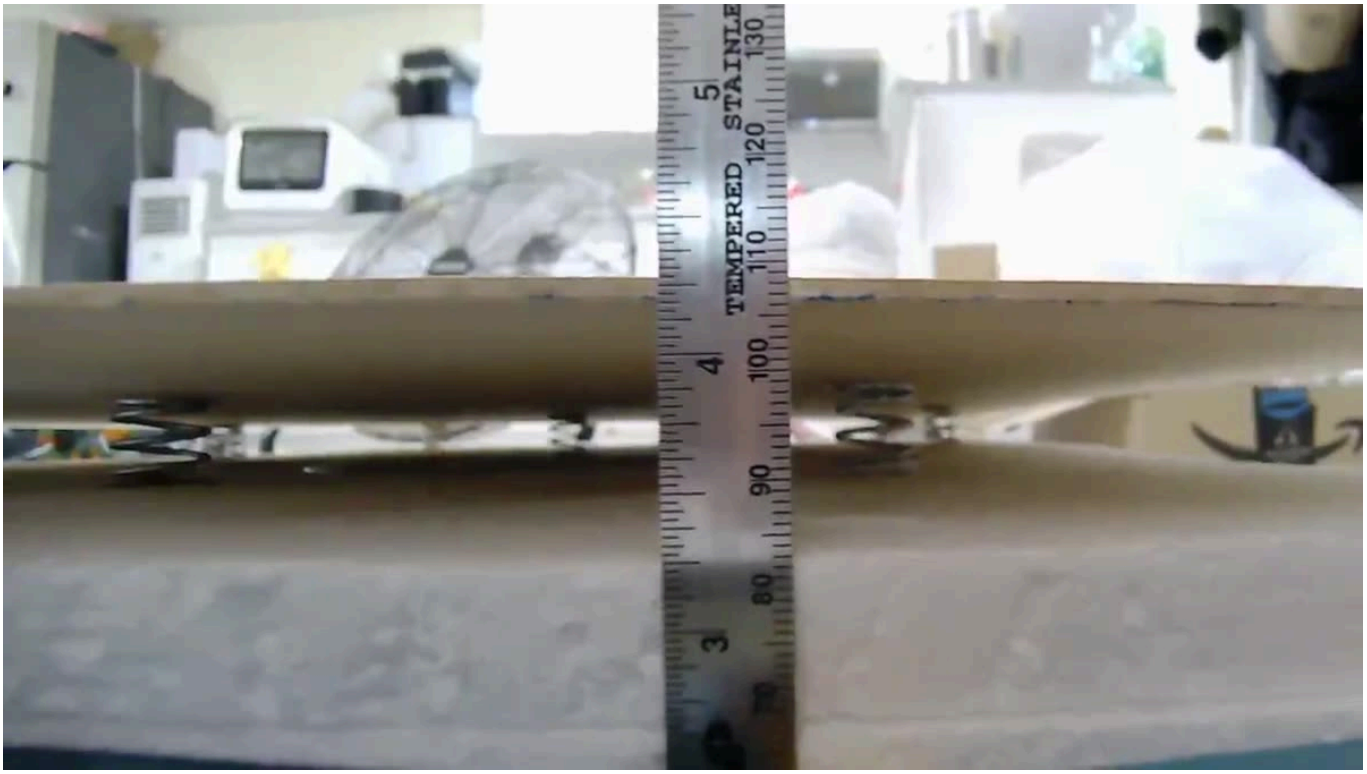
Curve	Clip time	Length
fast_1	17:32:16	7.6 s
fast_2	17:34:29	4.9 s
fast_3	17:34:56	4.4 s
slow_1	17:35:18	4.4 s
slow_2	17:35:29	3.3 s
slow_3	17:35:37	4.4 s



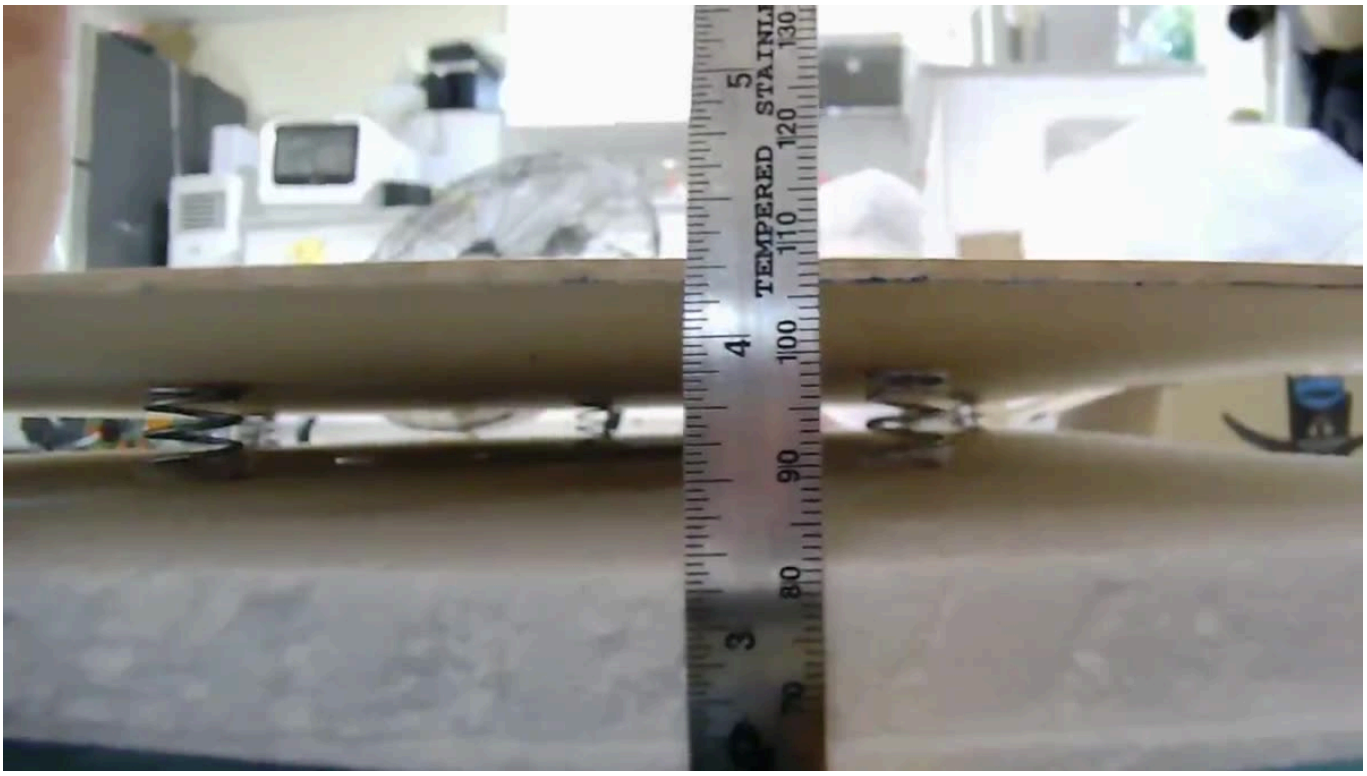
fast_1 recording *(video — still frame; play it in the online doc)*



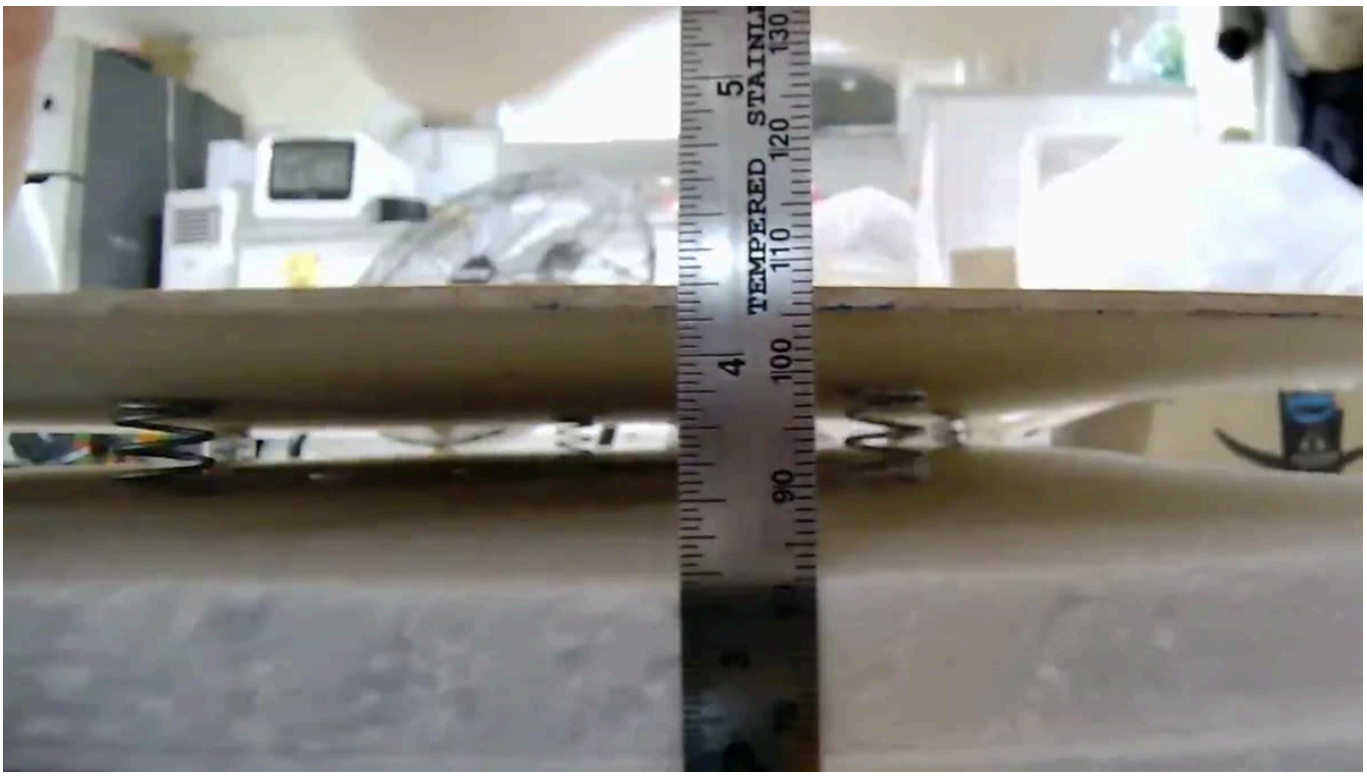
fast_2 recording *(video — still frame; play it in the online doc)*



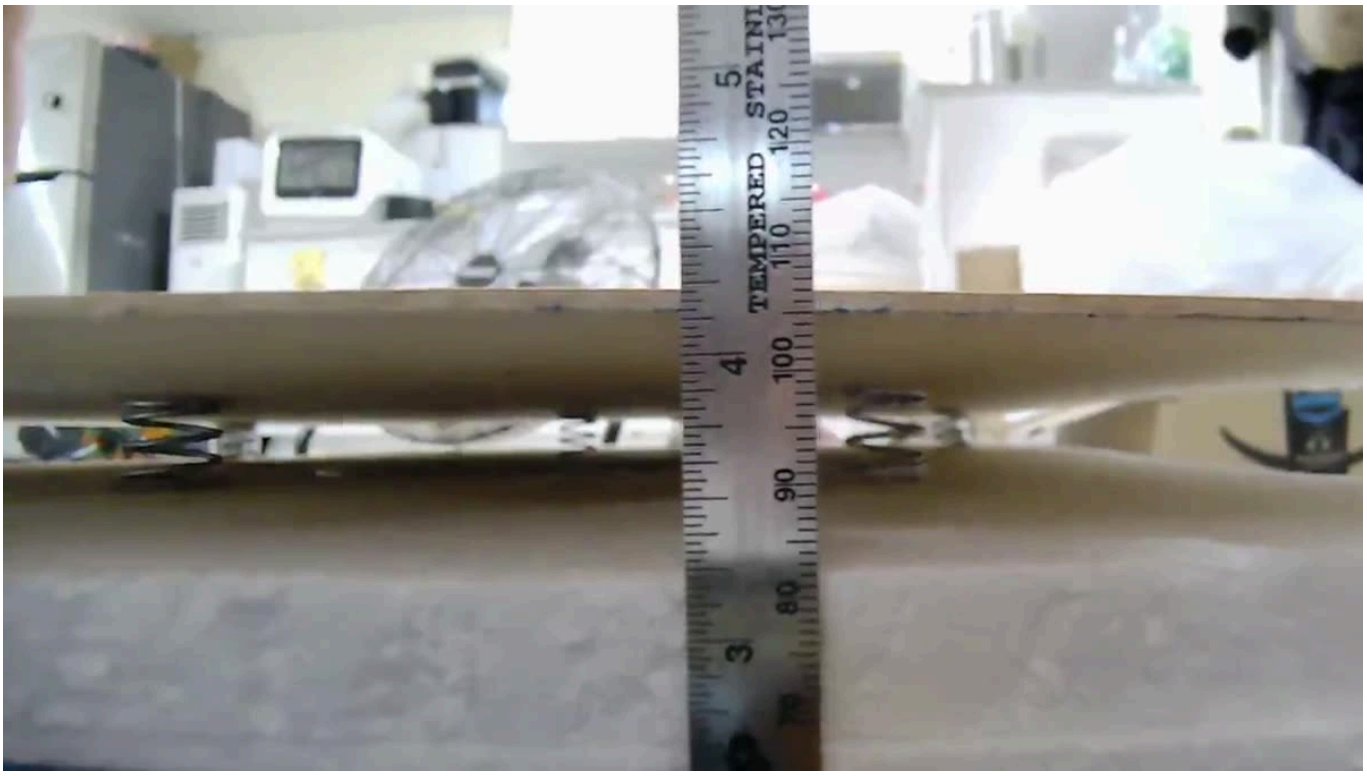
fast_3 recording *(video — still frame; play it in the online doc)*



slow_1 recording *(video — still frame; play it in the online doc)*



slow_2 recording *(video — still frame; play it in the online doc)*



slow_3 recording *(video — still frame; play it in the online doc)*

Open question: confirm the clip-to-curve mapping (assumed by recording order), and how plate position was read from the 120 fps frames into 0.01 s steps.

Simulation

An event-driven Python simulation (July 2026) screened impactor mass and spring stiffness against the measured step curves; it ranks 100 g + 0.5 N/mm as the first candidate to build. It checks mechanical feasibility only: it does not predict electrical output or disc cracking.

Question. With zero initial gap, can the floor's press-and-rebound make the impactor separate, fly free, and strike again?

Model (downward positive). While in contact the impactor moves with the floor, and the contact force is:

$$N(t) = m\,a_p(t) + k\,y_p(t)$$

The plate can only push, so $N < 0$ means separation. After separation the impactor is a free, undamped spring–mass oscillator, solved analytically to avoid integration drift:

$$m\,\ddot{y}_m + k\,y_m = 0, \qquad \omega = \sqrt{k/m}$$

A re-strike is the next root of $y_m(t) - y_p(t) = 0$ with closing relative velocity, updated with restitution e:

$$v_{m,\text{after}} = v_p - e\,(v_{m,\text{before}} - v_p)$$

Input data. Six measured curves (fast_1–3, slow_1–3), 32–44 points each at 100 Hz, converted from the Excel sheet to CSV (time_s, compression_mm). Floor velocity and acceleration come from spline interpolation and differentiation.

Analysis versions (these are pipeline revisions, not separate experiments):

Version	Change	Why
measured_v1	First run on measured curves; 6 trials × 3 smoothing × 3 masses × 4 stiffnesses	Many separations at the data's end turned out to be artifacts
measured_v2	Outside the data, set v = 0 and a = 0 instead of extending end slopes; flag early/tail events	160 separations, 56 valid, 99 suspect at the tail
measured_v3	Added raw vs. clamp preprocessing (negative end values set to 0) — 432 runs	Main analysis uses clamp; raw kept as a sensitivity check
v3 summary	Graded valid runs by impact speed, no rerun	Gives the candidate list below

Grading (216 clamp runs): a run counts only if separation is not in the first or last 20 ms and at least one re-strike occurs. Strong = relative impact speed ≥ 0.5 m/s. Result: 5 strong, 26 moderate (0.1–0.5 m/s), 16 borderline, 12 below threshold, 104 suspect, 53 no separation.

Curve	Mass	Spring k	Separation	Impacts	Max v_rel	Max energy	Role
fast_3	100 g	0.5 N/mm	~286 ms	2	1.17 m/s	68.3 mJ	Main candidate
fast_1	150 g	0.5 N/mm	~276 ms	1	1.11 m/s	91.7 mJ	Higher energy, harsher strike
fast_3	150 g	1.0 N/mm	~286 ms	2	0.96 m/s	69.3 mJ	Stiffer option
slow_1	150 g	0.5 N/mm	~68 ms	3	0.84 m/s	52.4 mJ	Early separation, needs separate explanation
fast_1	100 g	0.5 N/mm	~277 ms	1	0.64 m/s	20.4 mJ	Mild option

Fast steps produce the strong cases: a sharper rebound makes the m·a_p term pull N below zero. Main-analysis settings: preprocess = clamp, smooth = none, e = 0.5. Event times are reported to the millisecond, since the data is 100 Hz.

Limits: candidate range, not a final spring; impact energy is not electrical energy; no quantitative crack prediction. Next: a fine sweep of 80–150 g × 0.3–1.0 N/mm, then e = 0.3 and 0.7.

Spring order for physical tests

Three compression springs bracket the simulated 0.5–1.0 N/mm range (music wire / carbon spring steel, closed and ground ends, light oil or black oxide).

Spec	Rate	OD	Free length	Wire Ø	Solid height	Early target
A	0.517 N/mm (2.95 lbf/in)	42.85 mm	101.6 mm	2.32 mm	23.95 mm	0.5 N/mm, main
B	0.718 N/mm (4.10 lbf/in)	42.85 mm	101.6 mm	2.54 mm	27.03 mm	0.75 N/mm, stiffer
C	0.905 N/mm (5.17 lbf/in)	42.85 mm	101.6 mm	2.67 mm	27.79 mm	1.0 N/mm, stiffest

Tolerances asked of the supplier: rate ±10%, OD ±1 mm, free length ±2 mm. Inner clearance (rod reference) is 36.53 mm for A and 35.71 mm for B and C.

Planned: higher-speed motion capture

A 120 fps USB camera module (4K, wide-angle low-distortion lens, UVC) was selected to film plate motion at a finer time step than the current 0.01 s readings.

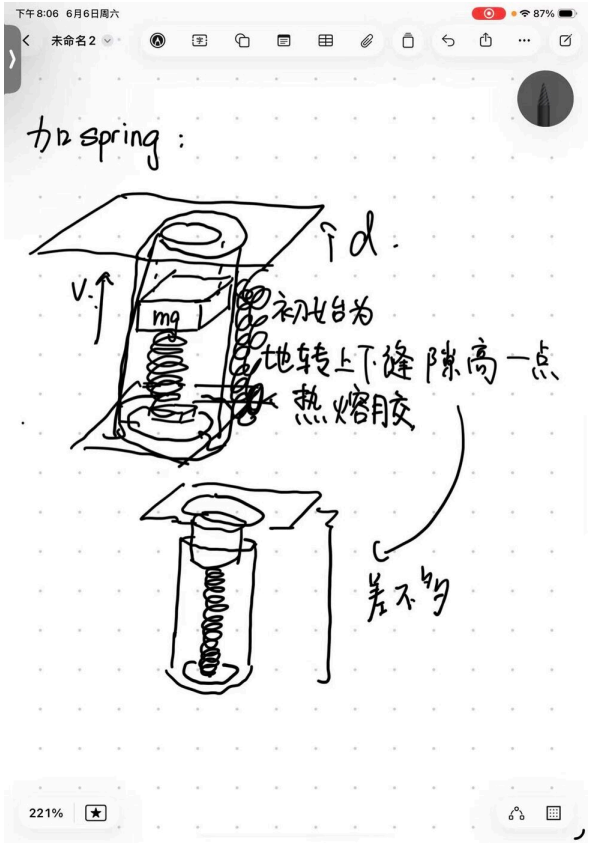


120 fps USB camera module selected for motion capture

Structure

The June 6 sketch sets the concept: each module is a tube standing between the top plate and the base. Inside it, an impactor (mass m) rides on a spring above a piezo disc. A support spring outside the tube, fixed with hot glue, carries the plate.

- Initial module height is set slightly taller than the gap between the top plate and base, so the impactor starts in contact (zero initial clearance).
- The lower sketch shows the alternative: a cap on a spring in the tube, sized about equal to the gap.
- The floor support system takes the body load and limits travel; the impact module does not carry the floor.



June 6 sketch: tube module with spring-mounted impactor (top) and capped tube about equal to the gap (bottom)

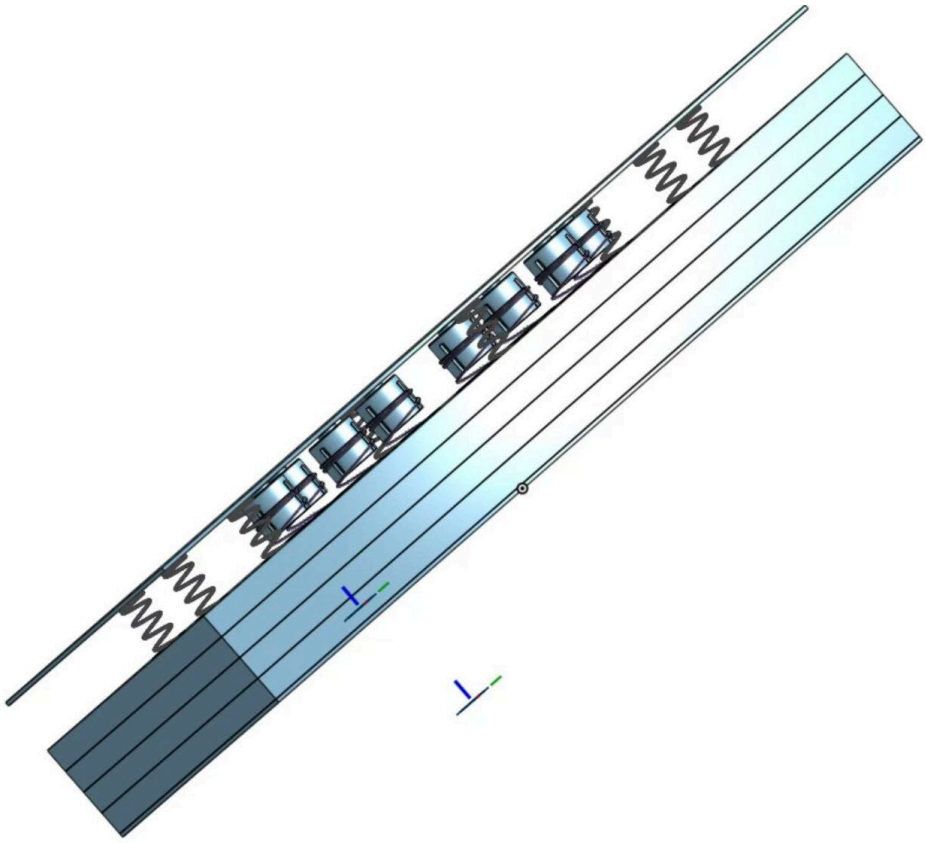
Built configuration (August 2026)

The built tile has 8 spring–impactor modules under one acrylic plate, with four corner springs that limit travel and return the plate.

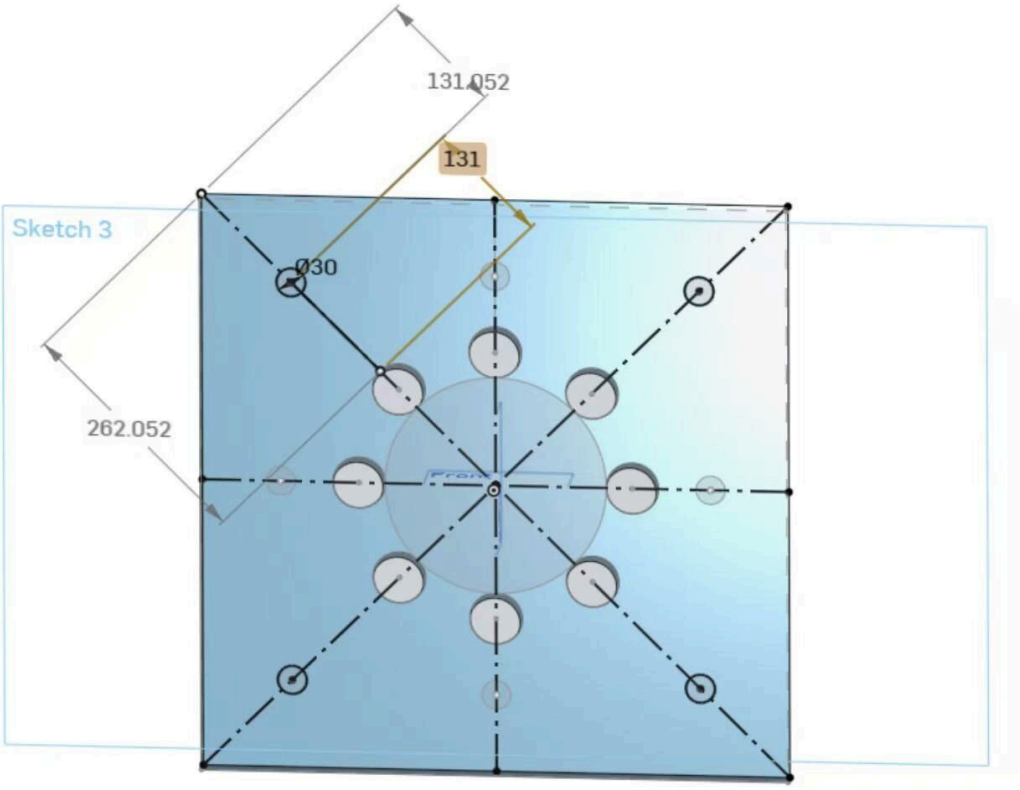
Part	Detail
Top plate	Acrylic, 455 × 610 × 2 mm; step near the center
Piezo discs	8, taped at two edges to the plate's underside, working in bending; all 8 in parallel → one bridge rectifier → 2.2 µF storage capacitor
Module stack (top to bottom)	Plate underside → upper sponge → piezo (ceramic face) → brass face → lower sponge → impactor tip → impactor → large spring → bottom support hole
Large spring	Sits loose in its hole, not glued or locked
Initial gap	Impactor tip just touches the lower sponge: $g_o = 0$, no added preload (to be checked per module after assembly)
Body	Four layers of EPS foam hold the modules and guide holes
Corner limit system	A spring + guide tube at each corner, glued to the plate, in the top three EPS layers; limits travel (design value ~25 mm, to be measured) and lifts the plate back

CAD (Onshape)

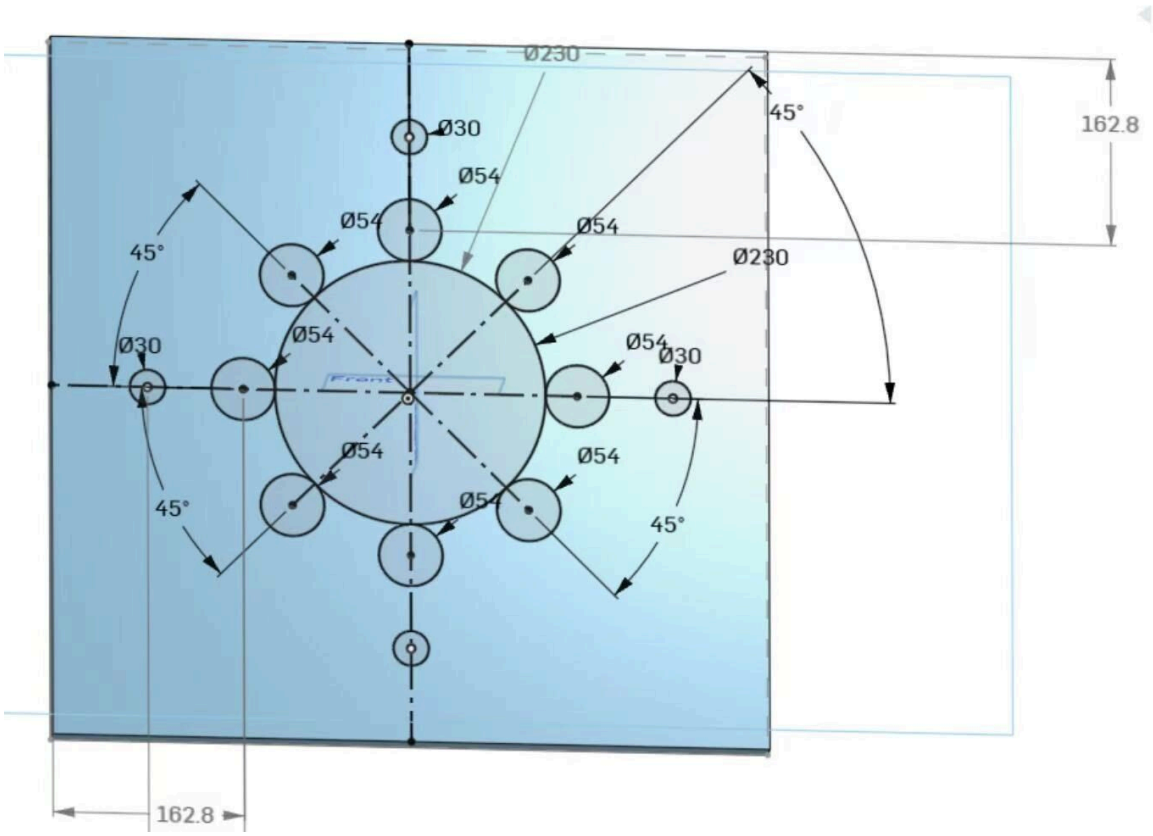
The layout places the 8 modules in Ø54 mm holes on a Ø230 mm circle, 45° apart, with Ø30 mm holes for the limit springs further out. The 3D model shows 8 limit springs: four at the corners and four at the edge midpoints.



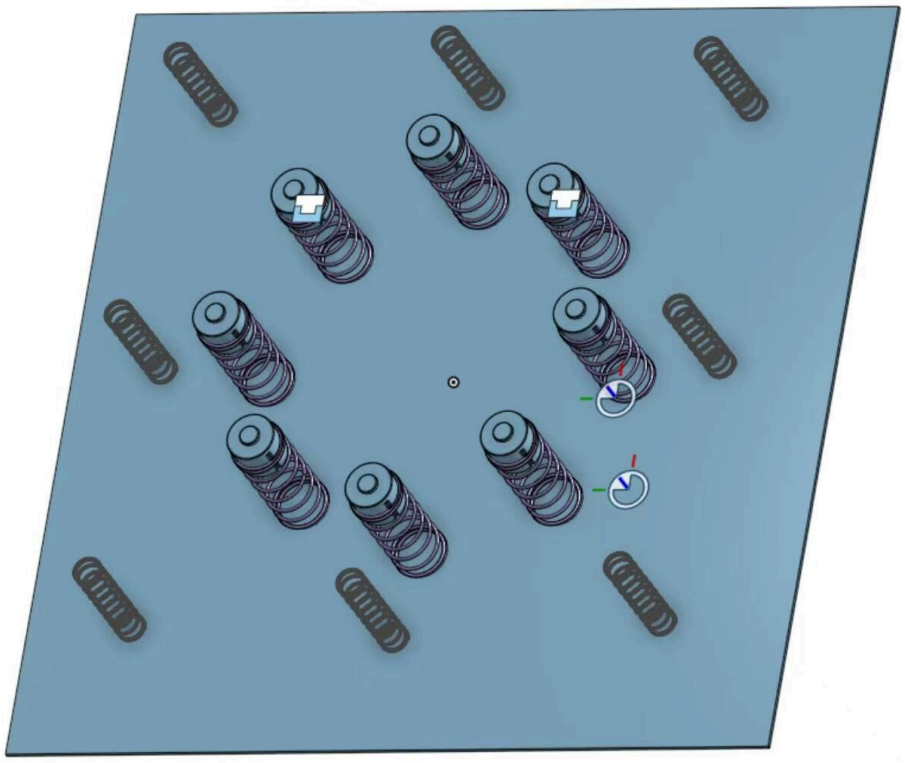
Version 2 CAD, section view: top plate, module springs and impactors, limit springs, foam layers



Version 2 layout sketch: Ø30 mm limit-spring holes on the diagonals (131 mm / 262 mm reference)



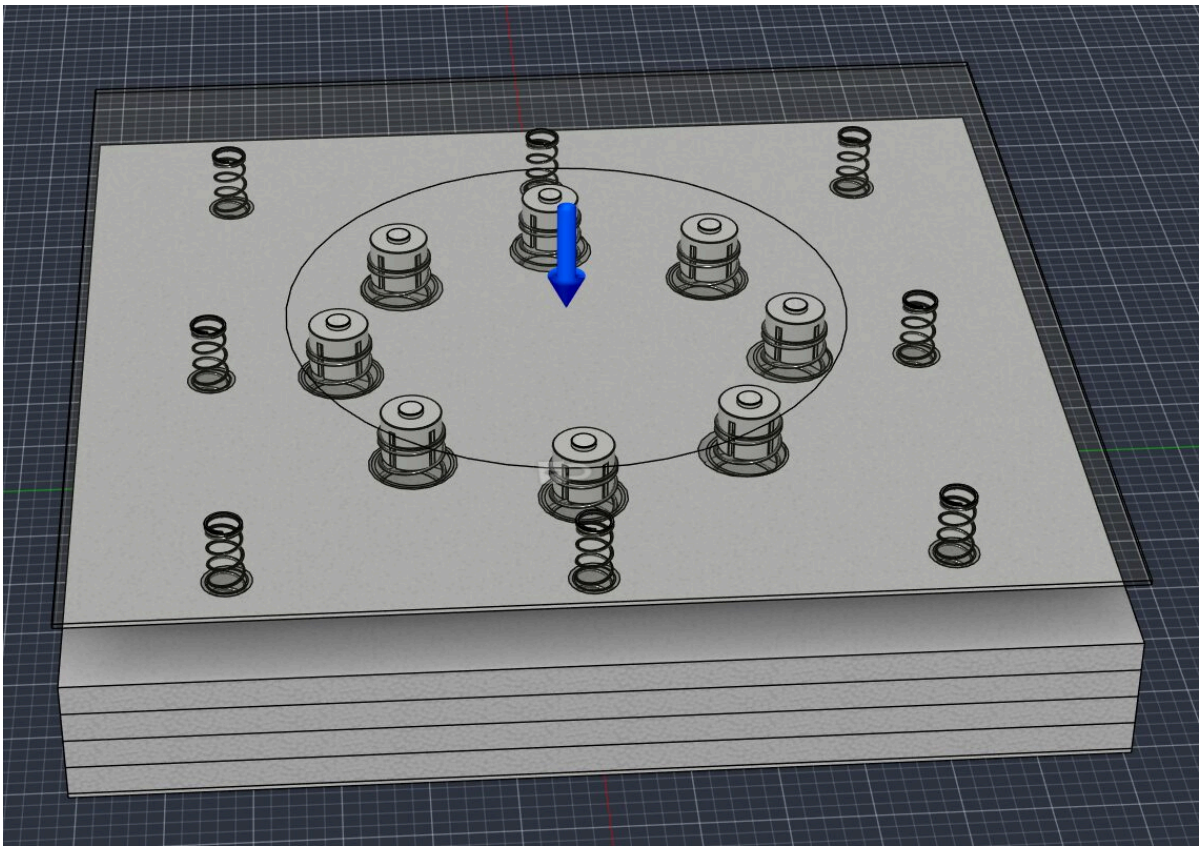
Version 2 layout sketch: 8 module holes Ø 54 mm on a Ø 230 mm circle at 45°, Ø 30 mm limit holes at 162.8 mm



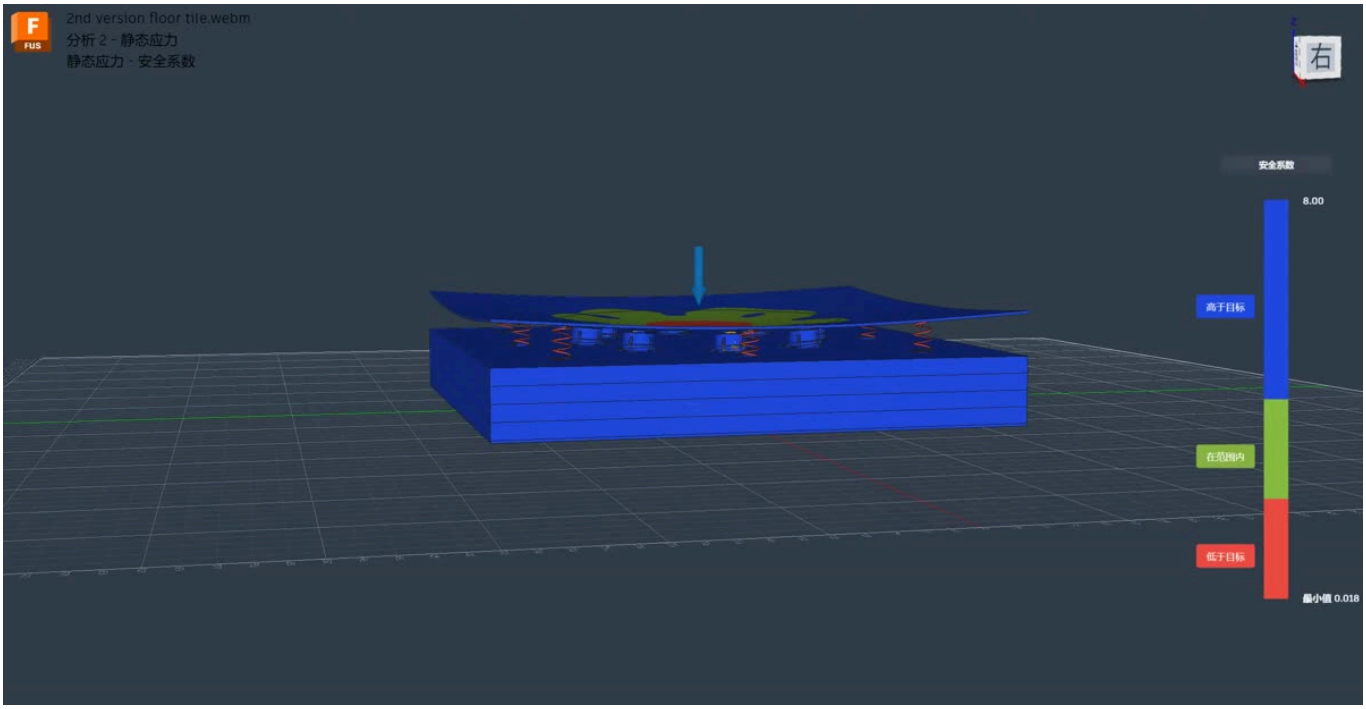
Version 2 3D model: 8 spring–impactor modules around the center and 8 limit springs around the edge

Fusion simulation

The full assembly (top plate, 8 modules, 8 limit springs, four foam layers) was rebuilt in Fusion and loaded at the plate center. The animation shows the plate deflecting under the load, colored by the safety-factor scale.



Version 2 Fusion model: center load (blue arrow) on the plate over 8 modules and 8 limit springs, four foam layers



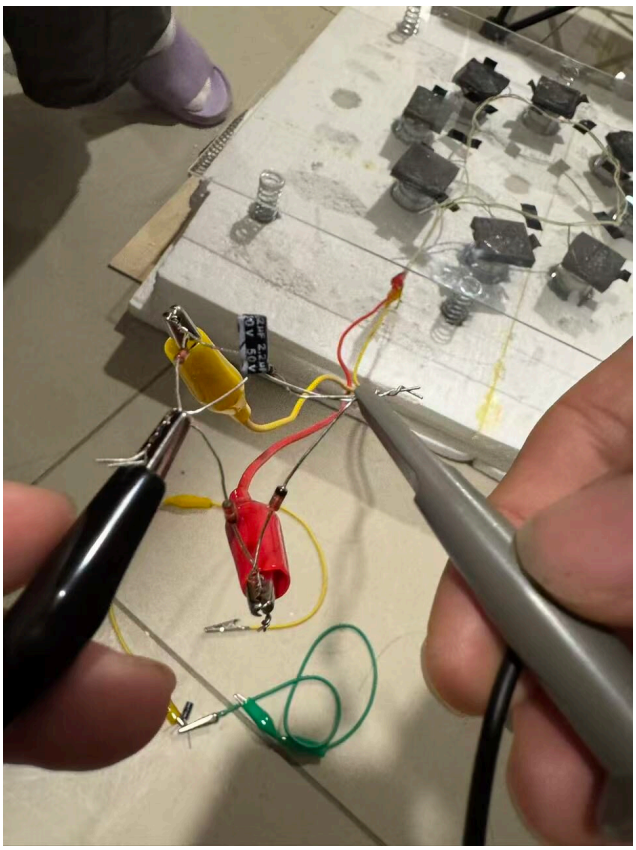
Version 2 Fusion simulation animation: plate deflection under a center load *(video — still frame; play it in the online doc)*

Build photos

The module housings were 3D printed as slotted cylinders. The assembled tile sits in white EPS foam under the acrylic plate, with the rectifier and 2.2 µF / 50 V capacitor wired at the edge and probed with the oscilloscope.



Version 2 module housings printing: slotted cylinders on the print bed



Version 2 assembled: piezo modules with sponge pads under the acrylic plate, limit springs in the foam, rectifier and 2.2 µF capacitor being probed

How one step makes charge:

1. The foot presses the plate; the sponge–piezo stack pushes the impactor down and compresses the large spring. The disc bends and gives the first pulse.
2. As the foot unloads, the corner springs lift the plate. When the contact force reaches zero, the impactor separates.
3. The impactor–spring oscillates freely, catches the rising plate, and strikes the lower sponge again: more pulses, each smaller.
4. The bridge turns all pulses into charging current. A pulse charges the capacitor only when $|V_{piezo}| > V_C + 2V_D$.

Test plan: New vs. Baseline

The test asks whether separation and re-strikes store more energy per step than a tile that keeps the impactor in contact. The main result is the net energy added to the storage capacitor per step:

$$\Delta E_{step} = \tfrac{1}{2}\,C_F\,(V_{after}^2 - V_{before}^2)$$

- **New:** the design above, free to separate and re-strike.

- **Primary Baseline:** same plate, discs, sponges, circuit and step position, with a light hold that keeps the impactor in contact, so the first press is unchanged but re-strikes are suppressed. The hold's own signal must stay below 5% of the main ΔV_C.
- **Measurements:** synchronized side video (plate and impactor motion, y_max, impacts, damping), open-circuit waveforms of the 8 discs, capacitor voltage before and after each step, optional resistor load; LED only as a demo.
- **Model checks from video:** damping ratio from peak decay, equivalent mass from the oscillation period, cross-checked against impactor mass + ⅓ spring mass.
- **Statistics:** at least 5 pilot pairs, randomized order within each pair, paired t-test with 95% confidence interval; sample size set from the pilot spread and a minimum meaningful difference chosen in advance.
- **Saving discs:** collect all quantities from the same steps in four batches (video + open circuit, capacitor pairs, continuous charging 10–20 steps, load/LED); inspect discs for cracks after each batch.

Open question: results of this test, and photos or CAD of the built tile.

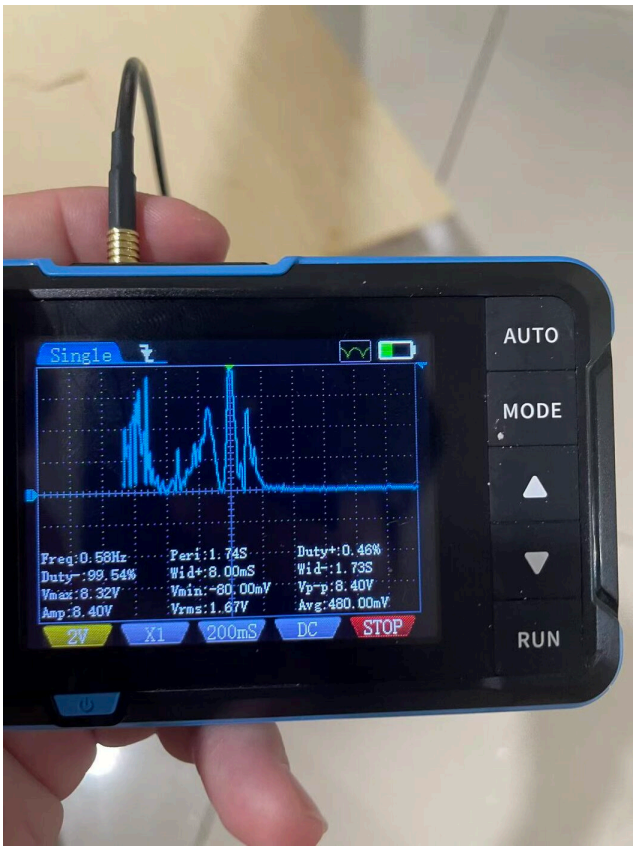
Testing and lessons learned

Open-circuit test

Five single steps on Version 2 gave 8.4–11.6 V peak-to-peak, and each step produced two or three separate pulse bursts instead of one. Scope probe set to X1, 200 ms/div.

Step	Scale	Max (V)	Min (V)	Vpp (V)
1	5 V/div	11.40	−0.20	11.60
2	2 V/div	8.32	−0.08	8.40
3	2 V/div	10.16	−0.08	10.24
4	2 V/div	9.84	−0.08	9.92
5	2 V/div	9.36	−0.08	9.44

- The trace is almost entirely positive, unlike Version 1's open-circuit pulses, which swung both ways.
- In steps 3 and 4 the tallest spikes reach the top of the screen at 2 V/div, so those maxima may be clipped.
- The later bursts, 0.3–0.6 s after the first, fit the press, the release, and possible impactor re-strikes; video would be needed to tell which is which.



Version 2 open circuit, step 2: three pulse bursts in one step (2 V/div, 200 ms/div)



Version 2 open circuit, step 5: first burst followed by two later bursts

Open question: number of discs connected, whether a rectifier was in line (the one-sided trace suggests it), and what led to Version 3.

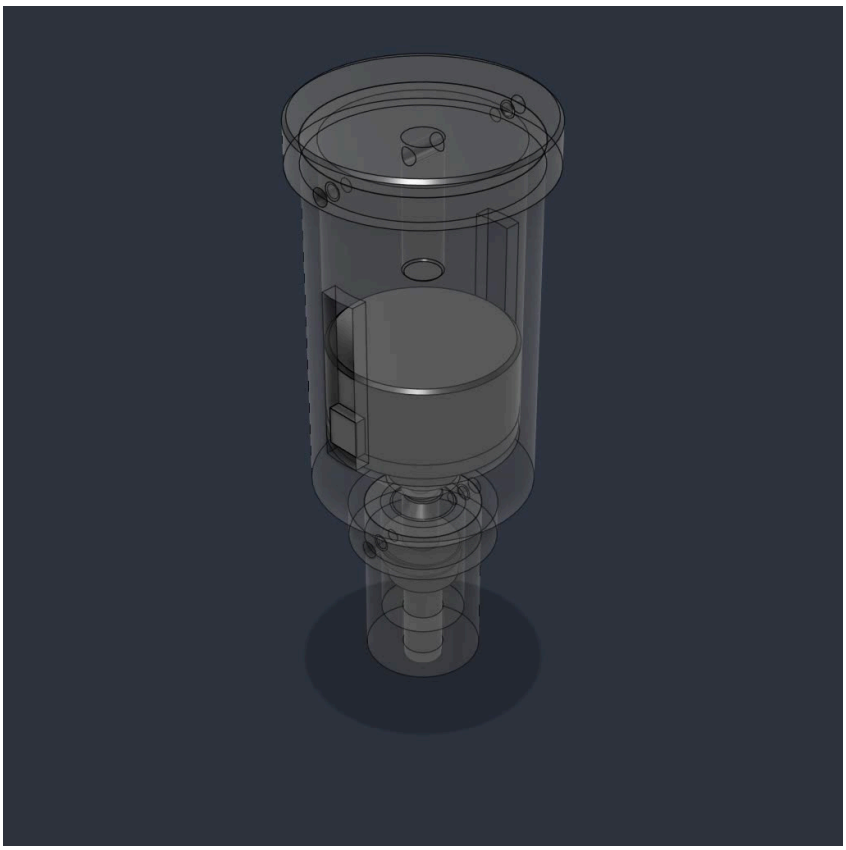
Version 3 — Center-punch tile

Version 3 copies an automatic center punch: a slow step stores energy in springs, a latch releases it mid-stroke, and a firing pin strikes the piezo at high speed. The strike force then depends on the released spring energy, not on how fast the person steps.

Mechanism

1. **Assembly.** The large upper spring is preloaded in the housing. The small lower spring sits in the yellow holder and rests on the red part of the green–red piece. The piezo disc sits inside the blue moving part, held from above by sponge and free to move only a little up and down.
2. **Step and store.** The step pushes the T-shaped input up relative to the housing. The green part and blue part move up, compressing both springs.
3. **Re-center and trigger.** The green part's guide reaches the ramp (throat) and is pushed back to center. When the thin green tip lines up with the small hole in the blue part, the latch releases. Triggering happens mid-stroke; the T-piece need not bottom out.
4. **Strike.** The T-piece and the green pin it backs act as a fixed anvil. The large spring drives the blue part and piezo down onto the green tip.

Key check: the blue part's lower stop must come after the tip contacts the piezo (s_stop > s_c). If the blue part lands first, the piezo only gets a slow push, not a punch.



Version 3 animation: one punch cycle — the input rises, compresses the springs, triggers and strikes (5 s) *(video — still frame; play it in the online doc)*

Hit-force model

The target is F_hit, the peak contact force of the green tip on the piezo. Speed at first contact, from spring energy released over the gap s_c:

$$\tfrac{1}{2} M v_c^2 = \tfrac{1}{2} k_b \left[x_{bt}^2 - (x_{bt} - s_c)^2\right] + M g s_c - W_{loss}$$

The contact path is treated as three springs in series (tip contact, piezo bending, sponge), measured together as one slope k_eq = ΔF / Δδ in a slow press test on the assembled stack:

$$\frac{1}{k_{eq}} = \frac{1}{k_c} + \frac{1}{k_p} + \frac{1}{k_f}$$

With the green end fixed (m_eff ≈ M), kinetic energy becomes contact deformation, giving the peak force:

$$F_{hit} = v_c\sqrt{m_{eff}\,k_{eq}}, \qquad \delta_{max} = v_c\sqrt{m_{eff}/k_{eq}}$$

Symbol	Meaning	How to get it
k_b	Large-spring rate	Compression test on the real spring
x_bt	Large-spring compression at trigger	Measure in the triggered state
s_c	Travel from trigger to first contact	CAD or caliper
M	Mass moving with the blue part	Scale
v_c	Speed just before contact	High-speed video if possible; else the energy formula with W_loss = 0 (upper bound)
k_eq	Contact stiffness of tip + piezo + sponge	Force gauge, slow press, slope of F vs. δ

Reporting rules: with measured v_c and k_eq, report an estimate in newtons; with v_c from the lossless formula, call it a theoretical upper bound. If δ_max exceeds the sponge's remaining travel, report both the soft and fully compressed stiffness cases. Body weight, spring static force or piezo voltage alone cannot give F_hit.

Structure

Open question: CAD and photos of the printed punch (about 15 cm tall, detachable 3D-printed parts), spring choices, and the firing-pin rework that made it trigger.

Testing

Open question: measured F_hit inputs, voltage output, and the upper-spring comparison.